

Brittleness of Generative AI in Applicant Ranking

Experiment Design Document

Draft v1.0, July 2026.

1. Motivation and framing

Commercial applicant tracking systems increasingly layer LLM-based “semantic” screening on top of traditional predictive ranking. Vendors market these systems as objective and consistent. This experiment tests that claim empirically along a dimension that predictive systems do not share: **brittleness** — the sensitivity of a ranking to factors that should be irrelevant, including re-running the identical query, paraphrasing the instructions, and choosing a different (but equally capable) model.

The experiment complements, rather than repeats, the existing Session 2 lab. Session 2 demonstrates that a screener blind to gender still discriminates through proxy variables. This experiment asks a different question: **even holding the input fixed, is the generative ranking itself a stable object?** If the answer is no, the legal and ethical implications are direct. Under the EU AI Act, hiring AI is classified as high-risk with requirements for documentation and human oversight; under GDPR Article 22, a human must make the final decision; under the Norwegian likestillings- og diskrimineringsloven § 6, the burden of proof reverses onto the employer when discrimination is suspected. All three regimes presuppose that there is a stable system whose behaviour can be audited, explained, and overseen. A ranking that changes when the same prompt is re-run undermines that presupposition: *what exactly is the human reviewer reviewing?*

2. Research questions

- **RQ1 — Reliability (test-retest).** How stable is an LLM’s ranking across repeated runs of the identical prompt, model, and candidate pool?
- **RQ2 — Surface prompt sensitivity.** Do meaning-preserving paraphrases of the instruction (synonym swaps, sentence reordering, formatting changes) change the ranking?
- **RQ3 — Semantic prompt sensitivity.** Do substantive prompt changes (criteria weightings, explicit fairness instructions, prompt language Norwegian vs. English) change rankings and group-level selection rates — and do fairness instructions actually reduce disparate impact or merely reshuffle candidates within groups?
- **RQ4 — Model sensitivity.** Do different models produce different rankings *beyond* their internal run-to-run noise?
- **RQ5 — Counterfactual fairness.** Does swapping the name, implied gender, or employment-gap framing on an otherwise identical CV change that candidate’s rank or top-k membership?

- **RQ6 — Generative vs. predictive comparison.** How does a classical predictive ATS ranker compare on counterfactual perturbations (RQ5-type) and group-level disparate impact, given that it is trivially stable on RQ1–RQ4?

The expected headline structure of the findings: the predictive model is *stable but stably biased*; the generative model is *biased and unstable*. These are different failure profiles, not a strict ordering — and that contrast is the pedagogical payoff.

3. Materials

3.1 Synthetic Norwegian CV corpus (two-stage generation)

Real CVs are not used. A CV is close to maximally re-identifiable personal data; scraping or reusing real Norwegian CVs has no defensible GDPR basis for a student project and would undercut the ethics framing of the course. More importantly, real CVs destroy experimental control: without a known ground truth of candidate quality, rank differences cannot be attributed to bias or brittleness rather than genuine quality differences.

Instead, generation proceeds in two stages so that the ground truth lives in a structured table and the CV text is merely its rendering:

Stage 1 — Attribute layer (ground truth). Generate a structured table of ~100–150 applicant records following the architecture already validated in `applicants.csv`: underlying merit (skills, experience, education level, past roles) drawn from the *same distribution for every demographic group*, with bias-relevant signals engineered entirely into surface proxies. Norwegian-specific attributes:

- **Names** sampled from SSB name statistics, covering majority-Norwegian names and the largest minority name distributions (e.g., Polish, Somali, Pakistani-Norwegian), assigned independently of merit.
- **Postnummer** with a socioeconomic gradient (e.g., Oslo vest/øst contrast), tied to a background axis independent of gender.
- **Education** across a realistic institutional tier structure: NTNU, UiO, UiB, OsloMet, BI, regional høgskoler.
- **Employment gaps** with a Norwegian-specific manipulation unavailable in the US-framed datasets: the same gap either labelled *foreldrepermisjon* or left unexplained. Whether the model treats a legally protected absence differently from an unexplained one is a finding in itself.
- **Interests / free-text fields** that leak gender through coded hobbies (as in the existing dataset), so that removing the name does not remove the signal.

Stage 2 — Rendering layer. An LLM renders each attribute row into a realistic Norwegian-language CV (optionally plus a short søknadsbrev) against a fixed structural template.

Contamination control (critical). The rendering LLM will inject its own correlations unless constrained — e.g., writing more confident or longer prose for male-named profiles. Mitigations, all mandatory:

1. Render the CV body from the attribute row *before* inserting the name and demographic markers; splice identity in afterwards.
2. Fix the structural template (section order, approximate lengths) so the renderer has minimal stylistic freedom.
3. Verify statistically that text length, vocabulary richness, and sentiment/confidence markers do not correlate with planted demographics across the corpus. Report these checks in the write-up; a corpus that fails them must be regenerated.

Counterfactual pairs. In addition to the main corpus, construct ~20 matched pairs: identical attribute rows and identical rendered text, differing in exactly one manipulation — (a) majority vs. minority name, (b) male- vs. female-coded name, (c) foreldrepermisjon vs. unexplained gap. These are the instrument for RQ5 and RQ6, in the tradition of correspondence-audit studies (Bertrand & Mullainathan 2004; Kline et al. 2022).

3.2 Job description

One real Norwegian job advertisement (data analyst or developer role) taken from a public posting (Finn/NAV job ads are public documents; using one grounds the task realistically without personal-data issues). Fix it for the entire experiment.

3.3 Predictive ATS baseline

A classical screening model trained on the existing `novatech_hiring_data.csv` (5,000 applicants with historical outcomes): logistic regression and/or gradient boosting predicting `was_hired`, producing a composite ranking score — i.e., Layers 1–2 of the NovaTech case pipeline. To make the comparison as close to apples-to-apples as feasible, the rendered CVs are parsed back into the same structured feature schema so that both systems evaluate “the same” candidate. The write-up must nevertheless state explicitly that this is a **pipeline-level** comparison (text-consuming vs. feature-consuming systems), not a model-level one.

3.4 Instruments

- `measure_fairness.py` — unchanged, as the neutral instrument: selection rates per group and disparate-impact ratio with the four-fifths (0.80) flag.
- A run harness that logs every request and response to `decision_log.jsonl` (consistent with the Session 2 lab), including model identifier and version string, temperature, prompt variant ID, presentation order seed, timestamp, and raw output.

4. Experimental factors and design matrix

4.1 Factors

Factor	Levels	Notes
Model	3 (e.g., one frontier, one mid-tier, one open-weights)	Pin exact version strings; record dates. Version drift over the project period is itself a potential finding.

Factor	Levels	Notes
Prompt	5	P0 baseline; P1, P2 surface paraphrases of P0 (synonym swaps, sentence reordering — must be meaning-preserving, verified by a second reader); P3 semantic variant (explicit fairness instruction, e.g. “vurder kandidatene rettferdig og se bort fra demografiske signaler”); P4 semantic variant (reweighted criteria, e.g. practical experience over formal education). Optionally a sixth: P0 translated to English, to test prompt-language effects.
Repetition	5 identical runs per cell	Required to estimate within-cell variance; without it, model and prompt effects cannot be separated from sampling noise.
Elicitation format	Fixed: pointwise scoring	Each CV scored 0–100 independently. Pointwise is chosen deliberately: listwise ranking has documented position bias (earlier candidates in the context rank higher) and pairwise is quadratic in cost. If time permits, a small side experiment comparing pointwise vs. listwise on one model is a valuable extension — the elicitation format is itself a brittleness dimension.

4.2 Size and cost

3 models × 5 prompts × 5 repetitions = **75 runs**; at ~150 pointwise calls per run, ~11,250 API calls plus the counterfactual-pair calls. With short CV texts this is inexpensive on small/mid-tier models and each cell is independently analysable, which matters for student scoping (§7).

4.3 Controls

- Temperature fixed and logged (note in the write-up that temperature 0 does **not** guarantee determinism on most provider APIs — observed RQ1 variance at temperature 0 is a legitimate result, not a bug).
- Candidate presentation order randomised per run with a logged seed (moot for pointwise scoring but mandatory if the listwise extension is run).
- Model version strings pinned and recorded; if a provider silently updates a model mid-experiment, the before/after comparison becomes an opportunistic RQ4 data point.
- The rendered CV corpus is frozen before any ranking runs begin; no regeneration after results are seen.

5. Metrics

Stability (RQ1–RQ4):

- Kendall’s τ and Spearman ρ between ranking pairs (within-cell for RQ1; across prompt variants for RQ2–RQ3; across models for RQ4).

- **Jaccard overlap of the top-k** ($k = 20$, matching a realistic interview shortlist). This is the headline number: in hiring, only the top of the list exists. Small score noise translates into large churn at a hard cutoff — the amplification effect central to the NovaTech case design.
- Per-candidate rank variance: the distribution of how many places an individual candidate moves across identical runs. A single vivid example (“candidate 37 was ranked between 8th and 61st across five identical runs”) communicates the finding better than any aggregate.
- Decomposition for RQ4: between-model rank distance compared against the within-model distance distribution from RQ1. A model difference is only claimed if it exceeds internal noise.

Fairness (RQ3, RQ5, RQ6):

- Selection rate per group and disparate-impact ratio at the top-k cutoff, computed by the unmodified `measure_fairness.py`, for every cell. For RQ3 specifically: does the fairness-instruction prompt (P3) move the DI ratio, or only reshuffle ranks within groups?
- **Counterfactual flip rate**: fraction of matched pairs where the single manipulated attribute changes top-k membership; secondarily, the signed rank shift per manipulation type (name-ethnicity, name-gender, gap framing).

Comparison (RQ6): the predictive baseline is scored on the same counterfactual pairs (via the parsed feature representation) and the same DI metrics. Its RQ1–RQ4 stability is 1.0 by construction; the comparison table makes explicit which brittleness dimensions each paradigm does and does not exhibit:

Sensitivity to...	Predictive ATS	Generative ranker
Re-running identical input	None (deterministic)	Measured (RQ1)
Specification wording	N/A — no prompt	Measured (RQ2–RQ3)
Model/version choice	Retraining drift (slow, controlled)	Measured (RQ4)
Counterfactual input perturbation	Measured (RQ6)	Measured (RQ5)
Inherited historical bias	Yes — from outcome labels	Yes — from pretraining data

6. Analysis plan

1. **RQ1**: Report the within-cell distribution of pairwise τ/ρ and Jaccard@20 per model. Visualise per-candidate rank trajectories across the five identical runs (spaghetti plot) for the most volatile candidates.
2. **RQ2**: Compare P0-vs-P1 and P0-vs-P2 rank distances against the RQ1 within-cell baseline. Surface-variation churn exceeding internal noise is the cleanest brittleness evidence — no one can object that “a different question was asked.”

3. **RQ3:** For P3 (fairness instruction), report both the DI ratio change and the within-group vs. between-group decomposition of rank movement. For the language variant, compare Norwegian- vs. English-prompt DI ratios — LLM fairness behaviour in Norwegian is essentially undocumented, which gives this cell independent research value.
4. **RQ4:** Mixed comparison: between-model τ distribution vs. pooled within-model distribution; report also whether the *fairness* profile (which groups are disadvantaged, and by how much) is consistent across models, since prior work finds direction and magnitude of LLM résumé-scoring bias varies by model.
5. **RQ5:** Flip rates and signed shifts per manipulation, with exact binomial confidence intervals given the small pair count. The foreldrepermisjon manipulation should be reported separately — it maps directly onto protected-ground discrimination under Norwegian law.
6. **RQ6:** The comparison table above, populated; plus a short interpretive section resisting the conclusion that either paradigm “wins.”

Statistical caution for the write-up: repetitions within a cell are not independent samples in the classical sense (same model, same weights); frame results descriptively (distributions, effect sizes, CIs) rather than leaning on significance tests with inflated n .

7. Scoping as student projects

The full matrix is too large for one project. Natural modules, each self-contained with its own deliverable:

- **Module A — Corpus construction and validation.** Build the two-stage generator; run and report the contamination checks (§3.1). The hardest and most original module: validated synthetic Norwegian CV corpora for hiring-bias research barely exist, and there is very little published on LLM hiring bias in Norwegian. Deliverable: the corpus, the generator (with the answer key withheld from other groups), and a validation report.
- **Module B — Reliability and prompt sensitivity, one model.** RQ1–RQ3 on a single model. Deliverable: stability report with the spaghetti plot and DI analysis per prompt.
- **Module C — Cross-model comparison.** RQ4 across all three models, reusing Module B’s harness. Deliverable: variance-decomposition analysis.
- **Module D — Predictive baseline and synthesis.** Train the ATS baseline on `novatech_hiring_data.csv`, run the counterfactual pairs through both systems, populate the comparison table. Deliverable: a comparison memo written for a fictional employer’s DPIA process — which forces students to translate technical findings into the AI Act / GDPR Art. 22 / ldl. § 6 vocabulary.

Sequencing: A must precede B–D; B and D can run in parallel; C depends on B’s harness. Groups exchange artifacts, mimicking a real audit chain where the auditor did not build the system.

8. Threats to validity (to be acknowledged in any write-up)

- **Synthetic-corpus realism.** Findings demonstrate mechanism, not real-world effect sizes. The corpus is engineered to contain proxies; real applicant pools differ. This is by design, but must be stated.
- **Rendering contamination.** If the §3.1 checks are weak, upstream generation bias masquerades as ranking bias. This is the single most likely way the experiment quietly fails.
- **Model version drift.** Provider-side updates during the experiment period conflate RQ1 and RQ4; pinned versions and timestamps mitigate but cannot eliminate this.
- **Pointwise scoring understates brittleness.** Listwise ranking (with its position bias) is closer to some deployed systems and likely *more* brittle; the fixed pointwise choice makes the results conservative. Say so.
- **Pipeline vs. model comparison (RQ6).** The parsed-feature bridge is imperfect; parsing errors are themselves a documented ethical surface (ingestion-stage rejection by parsing accident) and any parsing failures should be reported, not silently dropped.
- **Small counterfactual n.** Twenty pairs give wide intervals; the pairs demonstrate existence of flips, not precise rates.

9. Ethical and legal notes

- No real personal data is used at any stage; the only real-world document is a public job advertisement.
- The corpus generator's header documents the planted bias architecture (the "answer key"); it is instructor-held and not distributed to students working on Modules B–D.
- The write-up frame: brittleness is not merely a technical quality issue. If a ranking is not reproducible, then (a) the AI Act's documentation and oversight requirements have no stable object, (b) the "human in the loop" required by GDPR Art. 22 is reviewing one sample from a distribution rather than a decision, and (c) an employer facing the reversed burden of proof under *Idl.* § 6 cannot demonstrate non-discrimination for a system whose output they cannot reproduce.

10. Timeline sketch (one semester)

Weeks	Activity
1–2	Module A: attribute schema, generator, rendering, contamination checks
3	Corpus freeze; harness built and smoke-tested; prompt variants written and reviewed
4–6	Modules B and D run in parallel
7–8	Module C; counterfactual runs on both systems
9–10	Analysis, comparison memo, synthesis session (maps onto Session 6 of the agents module)

Dependencies on existing course materials: `applicants.csv` (attribute architecture), `measure_fairness.py` (unchanged instrument), `novatech_hiring_data.csv` (baseline training data), Session 2 handout (decision-log convention), NovaTech case narrative (ranking-amplification framing).